# RAHUL MITRA

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## EDUCATION

**National University of Singapore** **Aug 2023 – Jul 2027**
**Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)**

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.
- Relevant coursework: Software Engineering, Data Structures and Algorithms, Artificial Intelligence, 3D Computer Vision, Operations Research, Simulation, Statistics, Linear Algebra.

## WORK EXPERIENCE

**Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone)** **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine to run in any STMicroelectronics warehouse worldwide, ingesting each site's shelf coordinates, dimensions, and operating rules so one model adapts to any floor plan.

**Global Operational AI Systems & Data Engineer (Part-Time), Abbott Laboratories (Contract)** **Jun 2026 – Present**

- Deliver AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.
- Strengthen monitoring and process structure across healthcare supply chains with AI, protecting product availability for vulnerable patient groups.

**Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract)** **Jan 2026 – Jun 2026**

- Deployed a team-built APC simulator to production for live internal users on Azure App Service, with Docker packaging, access restrictions, and runtime and session hardening.

**Platform & Solutions Engineer (Sparks Citizen Developer), People's Association** **Sep 2025 – Sep 2026**

- Led most software development and deployment of a digital garden-plot balloting platform across admin and public Next.js frontends, a NestJS backend, and GIS mapping, now live in Tengah with national rollout planned.
- Integrated Singpass Login (OIDC) to secure resident sign-in and Myinfo to verify identity and address, keeping garden-plot balloting fair for 150 residents signed up at release.
- Surveyed garden sites into geospatial datasets for mapping, plot allocation, and site planning, and designed rules-based balloting across 4 control points (eligibility, conflict resolution, exceptions, admin oversight), keeping allocation auditable.
- Provisioned 10 AWS services with Terraform (ECS Fargate, Multi-AZ RDS Postgres with PostGIS, ALB behind WAFv2, Route 53, ACM, KMS, Secrets Manager, and a Rust auth gateway on Lambda and API Gateway), shipped via keyless GitHub Actions CI/CD on IAM OIDC across 5 repos with gated migrations, RDS snapshots, and auto-rollback.

## RESEARCH EXPERIENCE

**Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge)** **Jul 2026 – Oct 2026**

- Built vision systems turning raw camera feeds into perception-ready frames across 3 workstreams: camera ISP, super-resolution, and image restoration under real-world blur.

## PROJECTS

**Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams)** **Jan 2026**

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once.

## CO-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCES

**Elderly Care Volunteer, NTUC Health Active Ageing Centre** **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check health, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

## SKILLS AND CERTIFICATIONS

Technical: Solution architecture; cloud (AWS Fargate, Terraform, Azure, Docker, CI/CD); full-stack (React, TypeScript, FastAPI, Flask); Singpass Login & Myinfo (OIDC); RAG & AI agents; AI tools (Claude Code, ChatGPT, Codex, Gemini, ComfyUI); GIS mapping; accessibility; rapid prototyping
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery